Pediatric Genetic Disorder Prediction

22K	9K	0.79	0.71
Training Rows	Test Rows	T1 F1-Macro	T2 F1-Macro

Problem & Dataset

THE PROBLEM

- → Children present with overlapping symptoms
- → Family histories are often incomplete
- → Same disorder looks different patient to patient
- → Datasets are messy — missing values, noise, imbalance

DATASET OVERVIEW

44 Features

Demographics, genetics, vitals, labs, symptoms

3 Disorder Types

Mitochondrial · Single-gene · Multifactorial

9 Subtypes

Cystic fibrosis, Diabetes, Leigh syndrome + more

Metric

F1-Macro — equally weights all classes

Preprocessing Pipeline

01 Drop Noise Columns

Remove IDs, names, admin fields

02 Fix Data Types

Coerce numerics, strip whitespace

03 Handle Missing Values

Median / KNN imputation by missingness %

04 Outlier Treatment

IQR Winsorization on vital signs

05 Feature Engineering

Genetic risk score, symptom count, lab aggregates

06 Encode Categoricals

Binary map + label encoding

07 SMOTETomek Balancing

Oversample rare classes, clean borderline noise

08 Feature Selection

Mutual Information — drop near-zero score features

Modelling Approach

XGBoost

Excellent on tabular data, handles class weights, fast to tune

LightGBM

Very fast training, great on high-dimensional features, memory-efficient

CatBoost

Handles categoricals natively, robust to overfitting

SOFT-VOTING ENSEMBLE

Each model outputs class probability scores — the ensemble averages them. A stronger combined signal beats any single model, with better stability than stacking and far less training time.

3-fold cross-validation used as a sanity check before full training on all 22K rows.

Two Separate Models

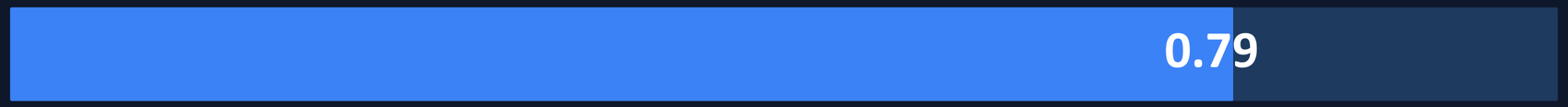
T1: 3-class Genetic Disorder
T2: 9-class Disorder Subclass

Each model weights features independently for its target.

Results

3-fold Cross-Validation · F1-Macro

Genetic Disorder (T1)



Disorder Subclass (T2)

4 in 5

Correct disorder category predictions across all 3 classes

9-Class

Subtype prediction — inherently harder; 0.71 is a strong result

Baseline

Both models far outperform majority-class naive baseline

Innovation Concept

Multi-Agent RAG-Based AI Family Doctor

User Query
Symptoms / reports via mobile or web

API Gateway
Secure, structured backend comms

Orchestrator Agent
Intent detection specialist routing

Specialist Agents (RAG)
Domain knowledge retrieval + LLM

Response Aggregation
Conflict resolution priority ranking

Safety & Validation
Red-flag checks disclaimers

Structured Response
Conditions + actions + follow-ups

RAG Pipeline per Agent:

Each specialist agent retrieves from its own domain knowledge base (clinical guidelines, research papers, patient history) instead of relying solely on the LLM's memory — ensuring accurate, grounded, up-to-date medical advice.

LLM **Language Models**

Natural language understanding and generation for clinical responses

RAG **Retrieval-Augmented Generation**

Live retrieval from domain knowledge bases for accuracy

MAS **Multi-Agent Collaboration**

Parallel specialist agents mimic real-world doctor consultation